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However the rate increases progressively, above a certain concentration of the substrate the curve flattened out. No further increase in rate occurs.

This shows that the enzyme is working at maximum velocity at this point. On the graph, this point of maximum velocity is shown as V_{max}.

8.6.9 Inhibitors of Enzyme

Certain substances present in the cells may react with the enzyme and lower the rate of reaction. These substances are called **inhibitors**. It is of two types **competitive and non-competitive** (Figure 8.33).

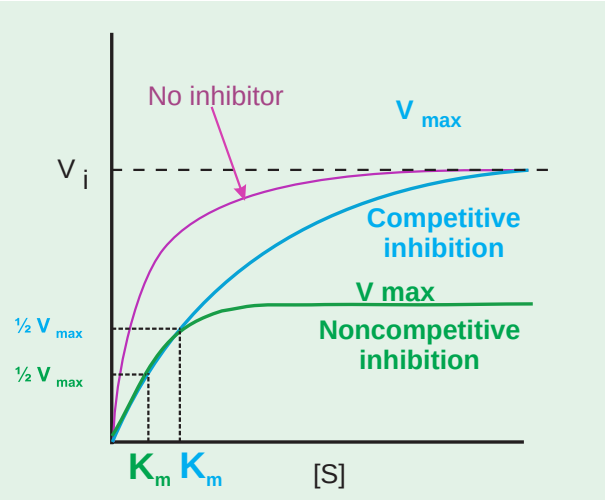


Figure 8.33: Enzyme inhibitors

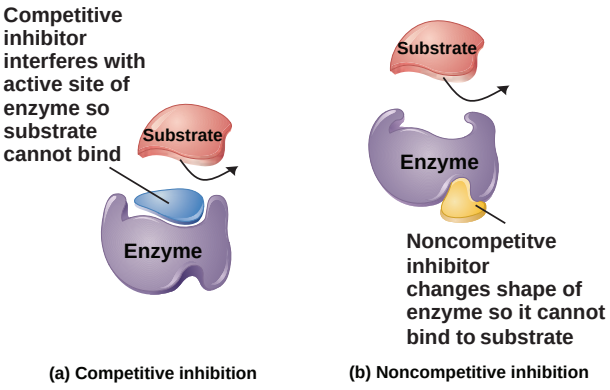


Figure 8.34: Action of Enzyme inhibitors

8.6.10 Competitive Inhibitor

Molecules that resemble the shape of the substrate and may compete to occupy the active site of enzyme are known as **competitive inhibitors**. For Example: the enzyme that catalyses the reaction between carbon di oxide and the CO_2 acceptor molecule in photosynthesis, known as **ribulose biphosphate carboxylase oxygenase (RUBISCO)** is competitively inhibited by **oxygen/carbon-di-oxide** in the chloroplast. The competitive inhibitor is **malonate** for **succinic dehydrogenase** (Figure 8.34).

8.6.11 Non-competitive Inhibitors

There are certain inhibitors which may be unlike the substrate molecule but still combines with the enzyme. This either blocks the attachment of the substrate to active site or change the shape so that it is unable to accept the substrate. For example the effect of the amino acids alanine on the enzyme pyruvate kinase in the final step of glycolysis.

Certain **non-reversible/irreversible inhibitors** bind tightly and permanently to an enzyme and destroy its catalytic properties entirely. These could also be termed as **poisons**. Example – **cyanide ions** which blocks **cytochrome oxidase** in terminal oxidation in cell aerobic respiration, the **nerve gas sarin** blocks a neurotransmitter in synapse transmission.

8.6.12 Allosteric Enzymes

They modify enzyme activity by causing a reversible change in the structure of the enzyme active site. This in turn affects the ability of the substrate to bind to the enzyme. Such compounds are called